

Mechanika Tuhých a Poddajných Telies

(MTPT)

P-05

KINEMATIKA

STATIKA - rovnováha silových účinkov pôsobiacich na hmotné objekty

KINEMATIKA - pohyb hmotných objektov v priestore a v čase - iba pohyb objektov bez uvažovania síl a hmotnosti

DYNAMIKA - pohyb hmotných objektov v priestore, v čase pri pôsobení síl a s uvažovaním hmotnosti objektov

Kinematika:

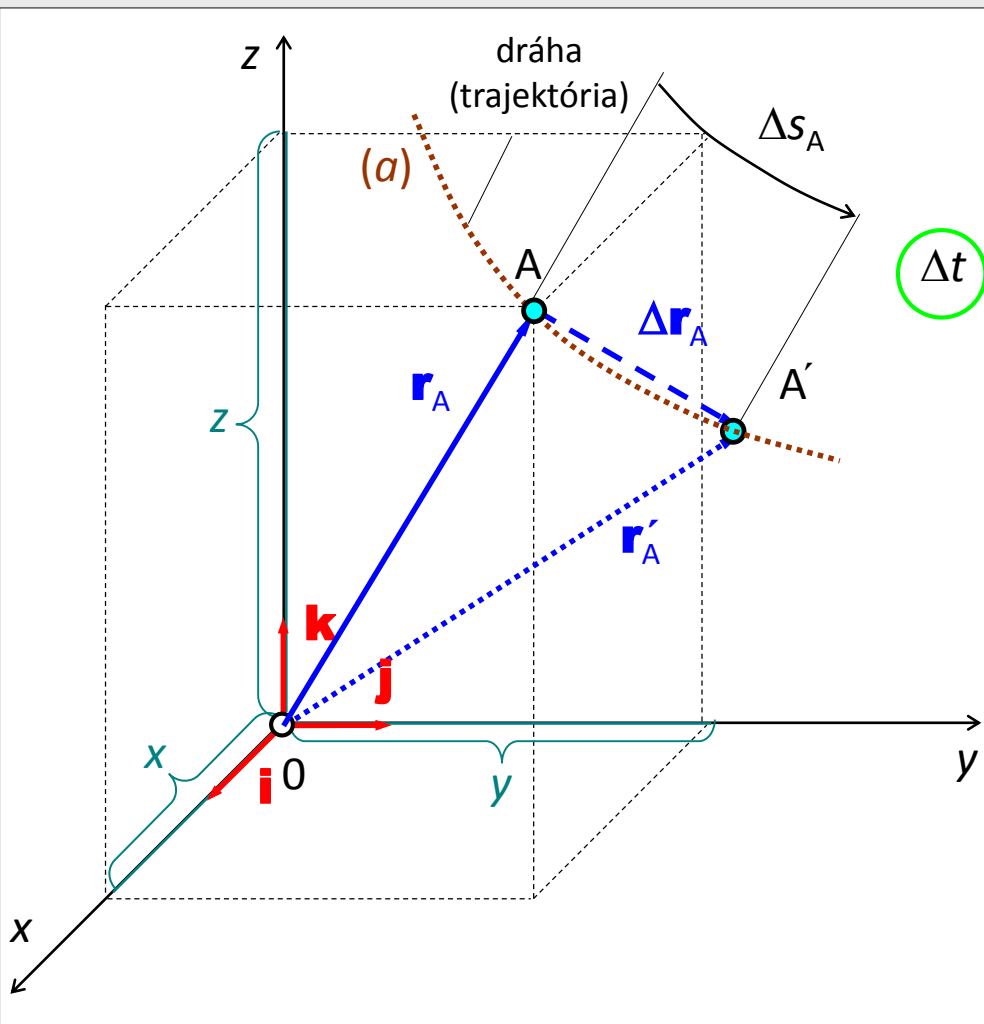
pohyb útvarov (bod, teleso) $\Rightarrow$ priestor + čas

Kinematika bodu

Kinematika telesa

Kinematika sústavy telies - mechanizmy

Kinematika bodu



Polohový vektor

$$\mathbf{r}_A = \mathbf{r}_A(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Vektor rychlosti

$$\mathbf{r}'_A = \mathbf{r}_A + \Delta \mathbf{r}_A$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}_A}{\Delta t} = \frac{d\mathbf{r}_A}{dt} = \dot{\mathbf{r}}_A = \mathbf{v}_A$$

$$\mathbf{v}_A = \frac{d\mathbf{r}_A}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}$$

$$\mathbf{v}_A = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$$

$$|\mathbf{v}_A| = v_A = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

Vektor zrýchlenia

$$\mathbf{v}'_A = \mathbf{v}_A + \Delta \mathbf{v}_A$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}_A}{\Delta t} = \frac{d\mathbf{v}_A}{dt} = \dot{\mathbf{v}}_A = \ddot{\mathbf{r}}_A = \mathbf{a}_A$$

$$\mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k}$$

$$\mathbf{a}_A = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} + \frac{d^2z}{dt^2}\mathbf{k} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

$$|\mathbf{a}_A| = a_A = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Uhlový pohyb

p_2 - pohyblivá priamka

p_1 - pevná priamka

Δt

$\Delta\varphi$

φ

$\mathbf{e}; |\mathbf{e}| = 1$

ω

ε

Uhol natočenia

$$\varphi = \varphi(t)$$

Vektor uhlovej rýchlosti

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta\varphi}{\Delta t} = \frac{d\varphi}{dt} = \dot{\varphi} = \omega$$

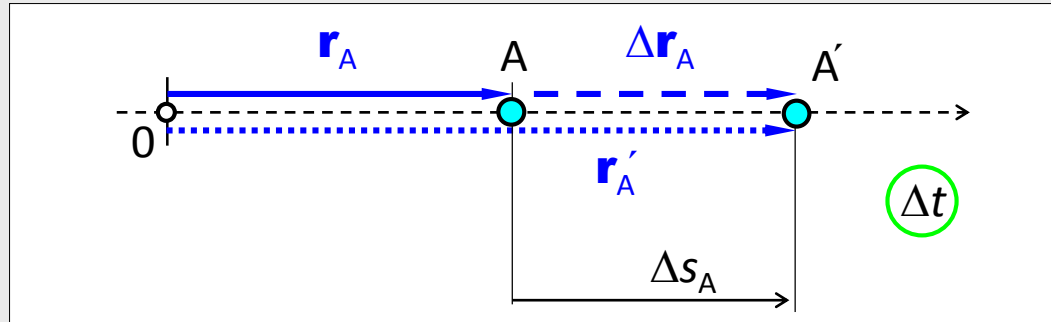
$$\boldsymbol{\omega} = \dot{\varphi} \mathbf{e} = \omega \mathbf{e}$$

Vektor uhlového zrýchlenia

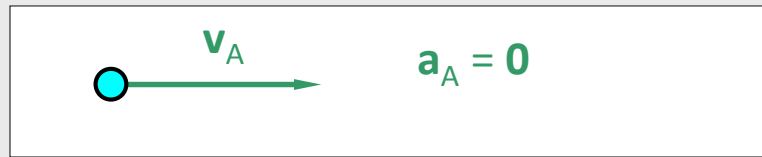
$$\lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \dot{\omega} = \frac{d\dot{\varphi}}{dt} = \frac{d^2\varphi}{dt^2} = \ddot{\varphi} = \varepsilon$$

$$\boldsymbol{\varepsilon} = \dot{\omega} \mathbf{e} = \ddot{\varphi} \mathbf{e} = \varepsilon \mathbf{e}$$

Pohyb bodu po priamke

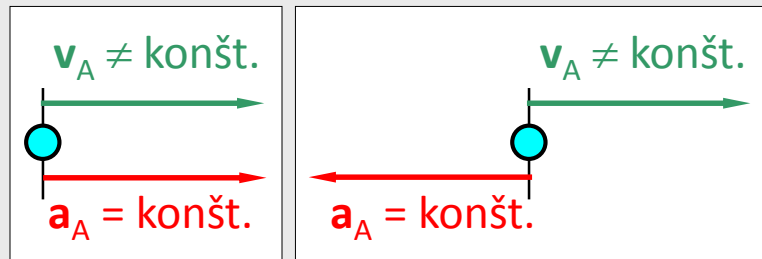


Rovnomerný pohyb



$$\mathbf{v}_A = \text{konšt.} \Rightarrow \mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} = 0$$

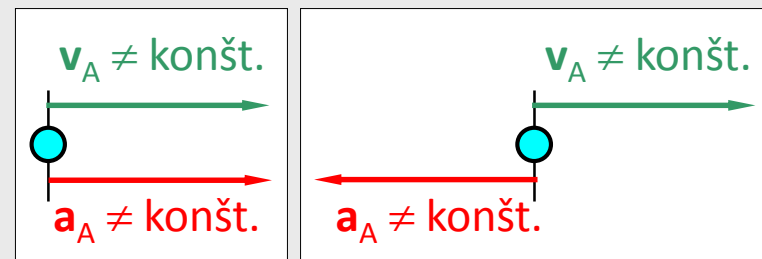
Rovnomerne zrýchlený/spomalený pohyb



$$\mathbf{v}_A \neq \text{konšt.} \Rightarrow \mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} = \text{konšt.} \quad \text{sign}(\mathbf{a}_A) = \text{sign}(\mathbf{v}_A)$$

$$\mathbf{v}_A \neq \text{konšt.} \Rightarrow \mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} = \text{konšt.} \quad \text{sign}(\mathbf{a}_A) \neq \text{sign}(\mathbf{v}_A)$$

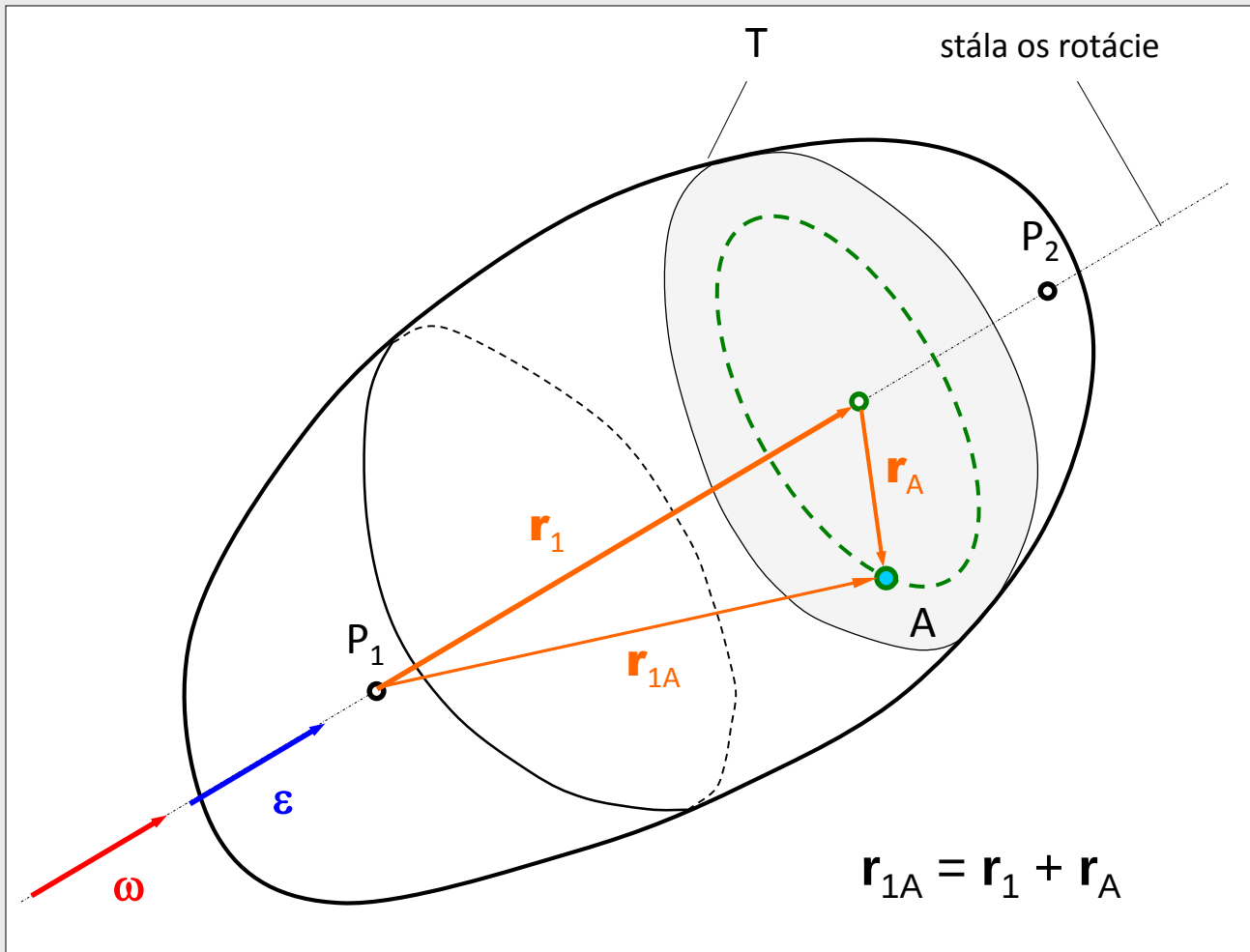
Nerovnomerne zrýchlený/spomalený pohyb



$$\mathbf{v}_A \neq \text{konšt.} \Rightarrow \mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} \neq \text{konšt.} \quad \text{sign}(\mathbf{a}_A) = \text{sign}(\mathbf{v}_A)$$

$$\mathbf{v}_A \neq \text{konšt.} \Rightarrow \mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} \neq \text{konšt.} \quad \text{sign}(\mathbf{a}_A) \neq \text{sign}(\mathbf{v}_A)$$

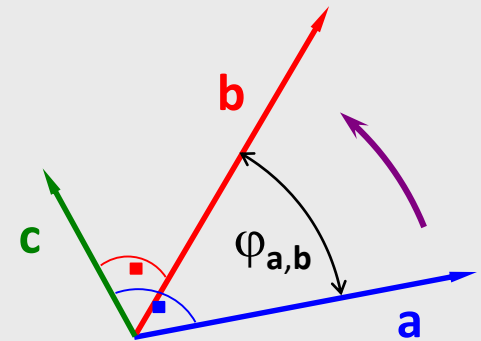
Pohyb bodu po kružnici



$P_1, P_2 \in T$

P_1, P_2 - nepohyblivé body

P_1, P_2 - ležia na stálej osi rotácie



$$\mathbf{c} = \mathbf{a} \times \mathbf{b}$$

$$|\mathbf{c}| = |\mathbf{a}| \cdot |\mathbf{b}| \cdot \sin \varphi_{a,b}$$

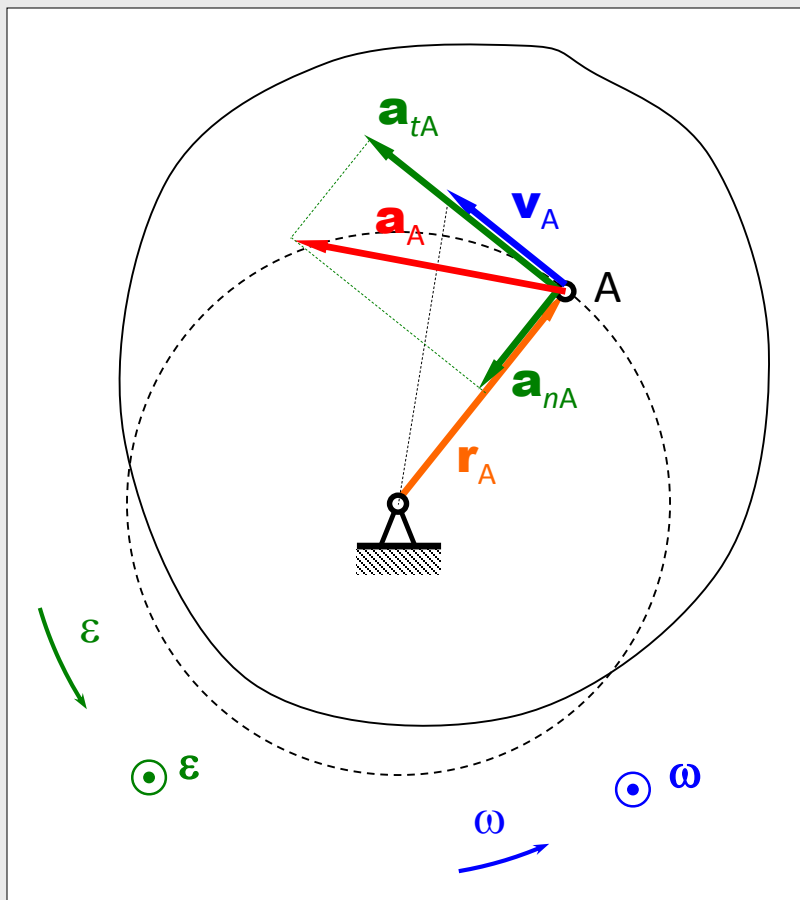
$$|\mathbf{c}| = a \cdot b \cdot \sin \varphi_{a,b}$$

$$\mathbf{c} \perp \mathbf{a} \quad \mathbf{c} \perp \mathbf{b}$$

$$\boldsymbol{\omega} \equiv \mathbf{r}_1 \Rightarrow \varphi_{\omega, r_1} = 0$$

Vektor rýchlosti $\mathbf{v}_A = \boldsymbol{\omega} \times \mathbf{r}_{1A} = \boldsymbol{\omega} \times (\mathbf{r}_1 + \mathbf{r}_A) = \underbrace{\boldsymbol{\omega} \times \mathbf{r}_1}_0 + \boldsymbol{\omega} \times \mathbf{r}_A = \boldsymbol{\omega} \times \mathbf{r}_A$

Pohyb bodu po kružnici



$$\mathbf{v}_A = \frac{d\mathbf{r}_A}{dt} = \boldsymbol{\omega} \times \mathbf{r}_A$$

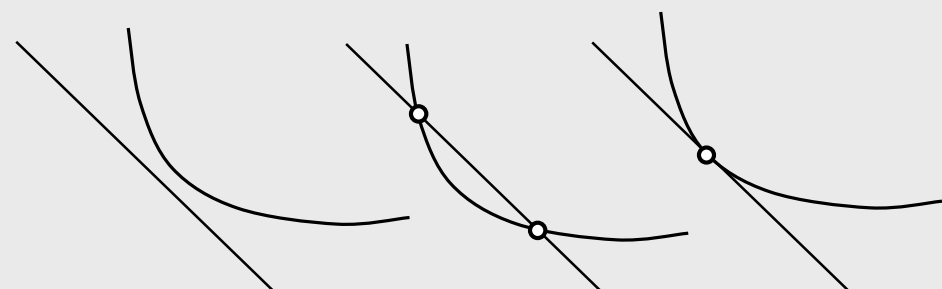
$$\mathbf{a}_A = \frac{d\mathbf{v}_A}{dt} = \frac{d(\boldsymbol{\omega} \times \mathbf{r}_A)}{dt} = \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r}_A + \boldsymbol{\omega} \times \frac{d\mathbf{r}_A}{dt}$$

$$\mathbf{a}_A = \boldsymbol{\epsilon} \times \mathbf{r}_A + \boldsymbol{\omega} \times \mathbf{v}_A = \boldsymbol{\epsilon} \times \mathbf{r}_A + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_A)$$

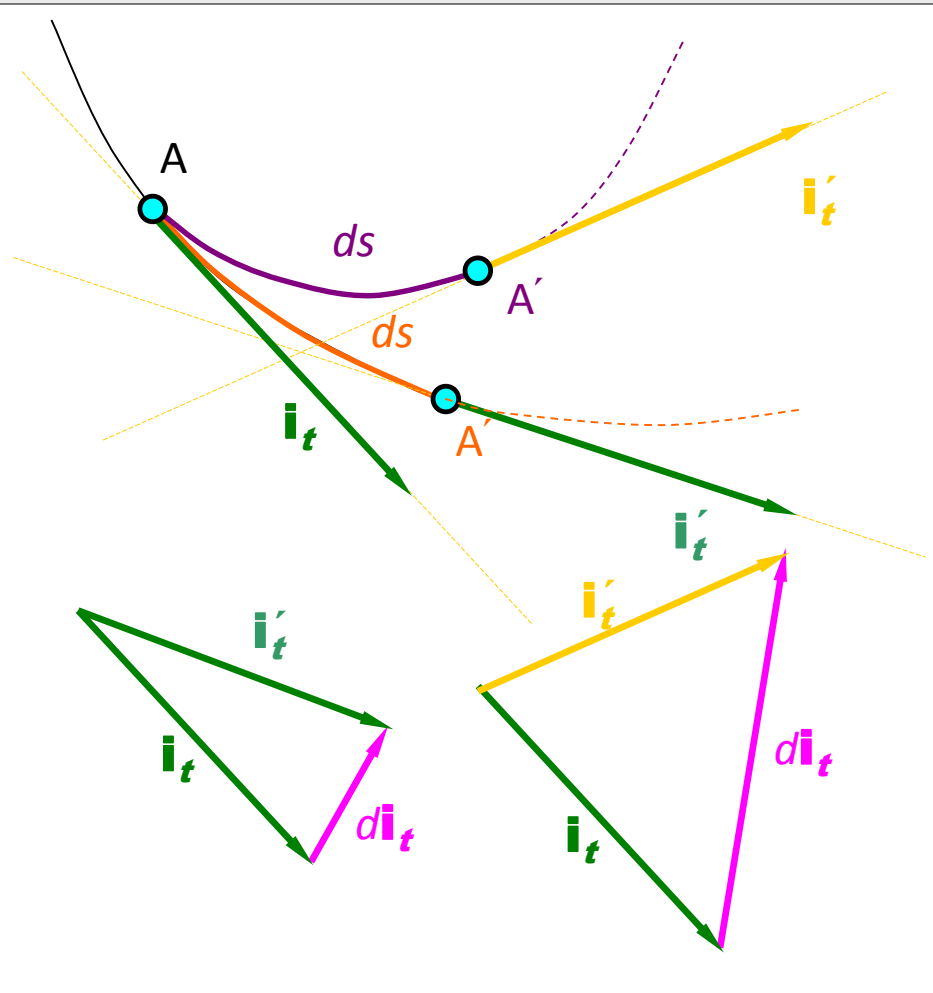
$$\mathbf{a}_A = \boldsymbol{\epsilon} \times \mathbf{r}_A + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_A) = \mathbf{a}_{tA} + \mathbf{a}_{nA}$$

$$\mathbf{a}_{tA} = \boldsymbol{\epsilon} \times \mathbf{r}_A$$

$$\mathbf{a}_{nA} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_A)$$

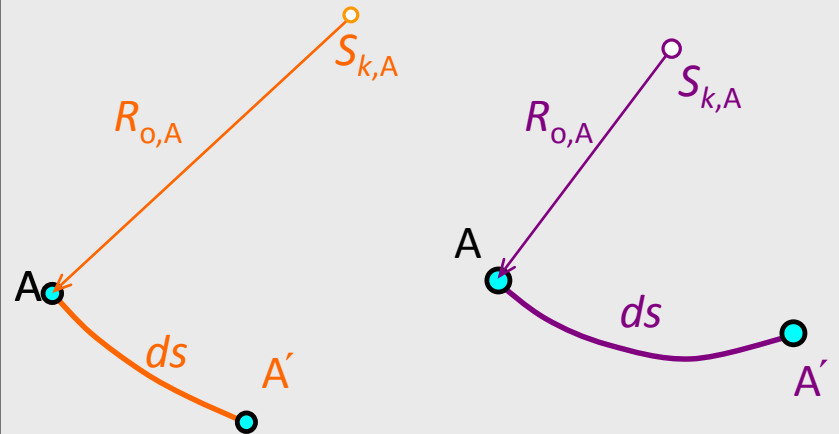


Pohyb bodu po všeobecnej krivke



$$\mathbf{K} = |\mathbf{K}| \mathbf{j}_n = K \cdot \mathbf{j}_n = \frac{1}{R_o} \mathbf{j}_n$$

$|\mathbf{K}|$ veľkosť vektora flexnej krivosti



$$\mathbf{K} = \frac{d\mathbf{i}_t}{ds} \quad \text{vektor flexnej krivosti}$$

$$d\mathbf{i}_t \perp \mathbf{i}_t \Rightarrow d\mathbf{i}_t \approx \mathbf{K} \perp \mathbf{i}_t$$

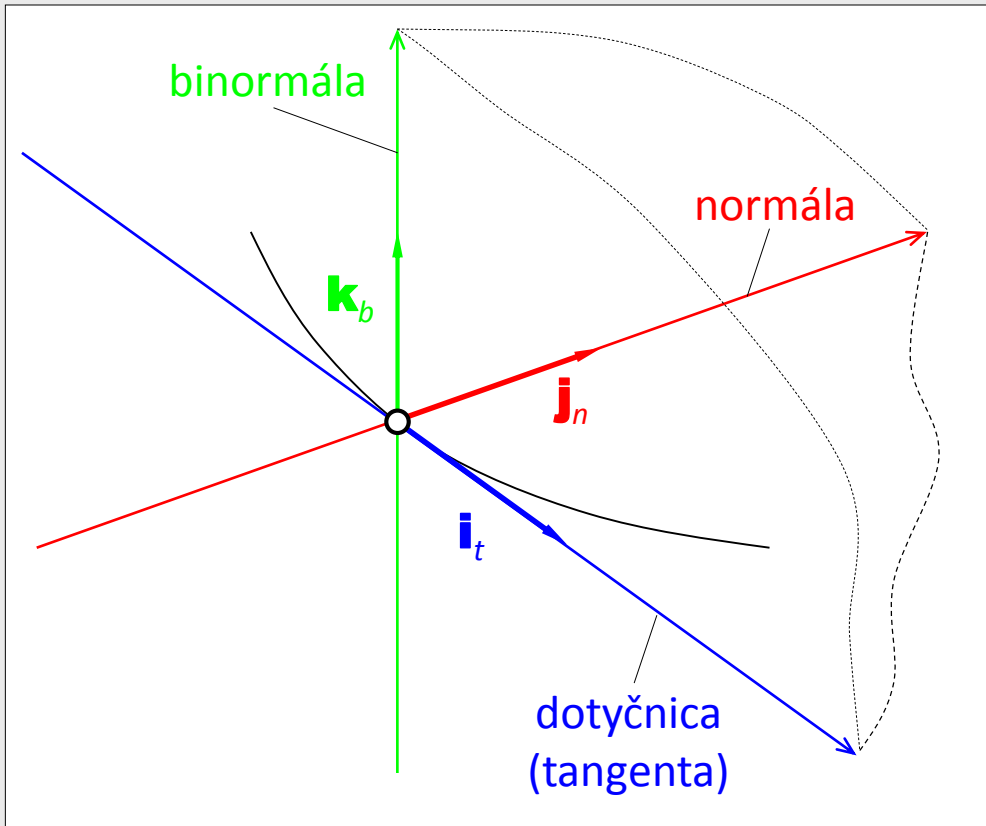
normála je kolmá na tangentu (dotyčnicu)

$\Rightarrow \mathbf{K}$ a $d\mathbf{i}_t$ majú smer normály
v smere normály je možné definovať
„jednotkový normálový vektor“

$$\mathbf{j}_n \perp \mathbf{i}_t$$

$$R_o = \frac{1}{|\mathbf{K}|} = \frac{1}{K} \quad \text{polomer krivosti dráhy}$$

Pohyb bodu po všeobecnej krivke



„jednotkový binormálový vektor“

$$\mathbf{k}_b = \mathbf{i}_t \times \mathbf{j}_n$$

jednotkový tangenciálny vektor

jednotkový normálový vektor

jednotkový binormálový vektor

} tvoria „Frenet-Serret sprievodný trojhran priestorovej krivky“

Pohyb bodu po všeobecnej krivke

Vektor rýchlosti $\mathbf{v} = v \cdot \mathbf{i}_t$

Vektor zrýchlenia

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d(v \cdot \mathbf{i}_t)}{dt} = \frac{dv}{dt} \cdot \mathbf{i}_t + v \cdot \underbrace{\frac{d\mathbf{i}_t}{dt}}$$

$$\frac{d\mathbf{i}_t}{dt} = \dot{s} \frac{d\mathbf{i}_t}{ds} = v \underbrace{\frac{d\mathbf{i}_t}{ds}}$$

$$\frac{d\mathbf{i}_t}{ds} = \mathbf{K} = |\mathbf{K}| \mathbf{j}_n = K \cdot \mathbf{j}_n = \frac{1}{R_o} \mathbf{j}_n$$

$$\mathbf{a} = \frac{dv}{dt} \cdot \mathbf{i}_t + \frac{v^2}{R_o} \cdot \mathbf{j}_n = a_t \mathbf{i}_t + a_n \mathbf{j}_n = \mathbf{a}_t + \mathbf{a}_n$$

$$a_t = \frac{dv}{dt}$$

$$a_n = \frac{v^2}{R_o}$$